The Hidden Cost of "Free": Open Educational Resources, Academic Rigor, and the Undermining of Higher Education Quality

The Issue

In light of the critical challenges surrounding student debt and the soaring costs of textbooks — often reaching $1,000 to $1,500 per academic year and $600 to $900 per semester depending on major — universities and colleges must confidently embrace the serious adoption of Open Educational Resources (OER). These invaluable resources, which include free, openly licensed textbooks, full courseware, question banks, simulations, and online course materials licensed under Creative Commons, are essential for easing the direct financial strain on students. This bold shift directly addresses the pressing and well-documented issue of textbook affordability, especially given that traditional commercial textbooks typically range from $200 to $400 each, with STEM, business, and health science titles often exceeding $300, and with new editions released every 2 to 3 years on a cycle that quickly renders used copies, rentals, and previous editions obsolete. For first-generation, low-income, and working students, these costs are not incidental. Surveys consistently find that 65-70% of students have skipped purchasing a required textbook due to cost, and nearly half report taking fewer courses or dropping a course specifically because of textbook costs. OER, when adopted intentionally, removes a significant financial barrier to access on day one of class.

While promoting OER is a commendable and necessary initiative for access and equity, it is equally important to scrutinize the implications for educational quality and instructional effectiveness. Many OER materials offer substantial value; leading repositories like OpenStax, Open Textbook Library, and LibreTexts now include peer-reviewed titles that rival commercial publishers in accuracy and scope. However, quality is uneven across the ecosystem. Some resources may not align with the rigorous standards, sequencing, and depth expected in specific academic programs. There are instances where these resources can lack comprehensive coverage of advanced or niche topics, fail to thoroughly explain complex concepts with the scaffolding students need, or provide inadequate practical examples, case studies, and scaffolded problem sets that are critical for skill development and mastery. Furthermore, the quality of illustrations, diagrams, charts, and multimedia elements — such as 3D anatomy models, interactive simulations, or adaptive homework — may not always meet the production standards, accessibility compliance, and consistency found in commercial materials backed by large editorial and design teams. The absence of integrated supplementary resources — such as vetted test banks, LMS-integrated quizzes, adaptive learning platforms, instructor slides, and lab manuals — can increase faculty preparation time and limit the effectiveness of teaching and learning experiences, with quality control processes that are often decentralized and less stringent than those for traditionally published content.

We must acknowledge the risks that come with relying solely on OER without appropriate review and support. If OER adoption is driven only by cost savings rather than by learning outcomes, students utilizing lower-quality or poorly matched resources might not achieve the depth of understanding, conceptual fluency, and procedural practice necessary for success in advanced studies, licensure exams, and professional settings. A student who saves $250 on an introductory biology text but enters organic chemistry or a nursing clinical without sufficient grounding in core mechanisms has not truly been served. This gap can result in downstream consequences: lower performance in prerequisite chains, increased DFW rates, and graduates feeling ill-equipped for the technical demands of their future careers and graduate academic challenges. Therefore, it is essential to champion a strategy that not only offers short-term cost savings but also measurably enhances educational quality, engagement, and fosters long-term student success, persistence, and career readiness.

Universities have a pivotal role in strategically balancing cost-saving measures with an unwavering commitment to maintaining high academic standards and faculty autonomy. This is not a binary choice between affordability and quality, but a design challenge that requires institutional investment. Open, evidence-based discussions about the selection, adaptation, and evaluation of OER and their impact on learning outcomes are critical to uncovering opportunities for innovation in teaching methodologies. By prioritizing both affordability and quality — through faculty review stipends, librarian and instructional designer support for curation and adaptation, student feedback loops, and alignment of OER with program-level learning objectives — we can ensure that students receive a superior education fully preparing them for their professional journeys. This approach not only drives improved student outcomes, retention, and timely completion, but also reinforces the institution's dedication to providing exceptional, equitable education while responsibly tackling the broader crisis of college affordability. Together, we are poised to navigate this transformative era in educational resources with confidence and a clear focus on the shared goal of student success.

The Textbook Affordability Crisis and Institutional Response: Assessing the Problem and Evaluating the Effectiveness of Proposed Solutions

Understanding the Scope of the Crisis

Rising commercial textbook costs create a substantial barrier to student achievement and widespread dissatisfaction. The U.S. Government Accountability Office (2013) reported an 82% increase in textbook prices from 2002 to 2012, far outpacing the 28% inflation rate. The College Board (2021) notes that students at public four-year colleges spend about $1,240 annually on books and supplies, while community college students spend around $1,460. Certain STEM textbooks cost between $200 and $400 or more (Senack, 2016). These financial burdens go beyond economic hardship and negatively affect academic performance. Students who cannot afford required materials are often less prepared, leading to decreased engagement, participation, and assignment completion. Over time, these challenges may reduce retention and graduation rates, especially among first-generation and underrepresented students. (Becker et al., 2023, pp. 1-20)

Beyond institutional initiatives, several policy interventions could address textbook affordability. State or federal governments might implement price caps to prevent excessive markups. Targeted grants or subsidies for faculty who develop or adapt high-quality Open Educational Resources (OER) could reduce student costs and promote instructional innovation. Policymakers could require publishers to separate digital access codes from print textbooks, increasing the availability of used books. Transparent textbook adoption guidelines would ensure cost considerations are part of curricular decisions. Additionally, partnerships with public consortia could facilitate bulk purchasing or licensing agreements, reducing prices for all students. ("Open Textbooks Pilot Program")

Students use various strategies to reduce textbook costs, including purchasing used books, renting, sharing with peers, or borrowing from libraries. Many also seek digital versions or free online resources. Faculty contribute by selecting affordable materials, adopting open resources, and advocating for departmental support. Prioritizing cost considerations and fostering interdepartmental collaboration improves student access to essential resources. University leadership plays a key role in setting priorities, allocating resources for sustainable textbook solutions, and promoting collaboration among departments and student services. Through these systemic approaches, institutional leaders can achieve lasting impact. (Walsh)

Research indicates that publishers frequently engage in practices that disadvantage students. New editions are released every two or three years with minimal changes, making it difficult to purchase used books (Hilton et al., 2013). Publishers also bundle textbooks with expensive, single-use online access codes that cannot be resold (Florida Virtual Campus, 2016). Furthermore, U.S. students often pay significantly more than international students for identical content (Jhangiani & Jhangiani, 2017).

Elevated textbook costs present significant challenges for students. In a survey of over 22,000 Florida college students, 37.6% did not purchase a required textbook due to cost (Florida State University, 2016). Consequently, 37.6% received a poor grade, 19.8% failed a course, and 47.6% enrolled in fewer classes. Colvard et al. (2018) found that textbook expenses disproportionately affect Pell Grant recipients, underrepresented minorities, and part-time students.

Analyzing University Motivations for Promoting OER

Universities often gain public relations and political benefits from Open Educational Resources (OER) initiatives, although these efforts rarely address the root causes of rising college costs. Henson (2017) found that most OER policies at 50 institutions were motivated by the pursuit of positive media coverage, deflection of criticism regarding tuition increases, responses to political pressure, attraction of cost-conscious students, and demonstrations of innovation to accreditors and stakeholders. However, some institutions exhibit a genuine commitment to student success by thoughtfully implementing OER initiatives and achieving positive outcomes in affordability and instructional quality. In these cases, effective OER leadership is essential. Certain presidents and provosts have established OER task forces, incorporated faculty and student input into decision-making, and allocated resources such as funding or grants for adapting open materials. Effective leaders set clear objectives that connect OER adoption to student learning, equity, and faculty innovation, rather than focusing exclusively on cost savings. They also promote transparency by sharing data on OER impacts and reporting both savings and ongoing investments.

Research and best practices suggest that effective OER policies include several essential elements: (1) transparent reporting of both costs and savings, (2) meaningful faculty and student involvement in OER selection and implementation, (3) clear resource allocation, such as funding or release time for faculty, (4) guidelines for evaluating educational quality and learning outcomes, and (5) regular public updates on progress and impacts. Incorporating these components enables institutions and policymakers to design OER initiatives that promote affordability, teaching quality, and student success. These practices demonstrate that universities can align OER initiatives with their core educational values.

Available data show that OER initiatives often improve institutional image without addressing fundamental cost issues. Textbooks represent only 2-4% of total college expenses, while tuition and fees account for 50-70% (College Board, 2021). Other significant costs include on-campus housing and mandatory fees, which often increase faster than textbook prices and make up a larger share of student budgets. Although institutions may report that OER saves students $500-$1,000 annually, tuition typically rises by $1,000-$3,000 each year. As a result, students may face higher overall costs despite reported savings. Between 2015 and 2020, as OER adoption expanded, average tuition at public four-year colleges increased by 20-35% after adjusting for inflation, far exceeding textbook savings (SHEEO, 2020).

OER initiatives let institutions project a cost-conscious image without making substantive budget reforms. Instead of reducing administrative positions, which grew 60% faster than enrollment between 2000 and 2020 (Ginsberg, 2011), cutting executive pay, or limiting spending on amenities, facilities, athletics, and marketing, institutions often mandate the use of "free" textbooks. This strategy often shifts costs to faculty, who must spend extra time locating, customizing, or creating OER materials without enough support or recognition. Such mandates may also reduce faculty autonomy in selecting course materials and designing curricula, especially when specific OER resources are required. For students, these policies can lead to diminished learning experiences. Addressing textbook costs is a politically convenient approach that avoids deeper discussions about institutional priorities. Sustainable change requires campus leadership to balance reducing student expenses with strong support for faculty workload and academic quality, ensuring reforms meet the needs of both educators and students. (Walsh)

To protect faculty time and autonomy, policymakers and institutions should implement explicit workload protections. These measures include compensating faculty for developing or adapting OER through stipends, course releases, or professional development funding. Institutions should clarify that participation in OER initiatives is voluntary unless otherwise negotiated and should recognize OER-related work in promotion and tenure evaluations. Collaborative guidelines and shared decision-making between administration and faculty senates can help address potential challenges. These strategies support sustainable OER adoption while maintaining instructional quality and faculty well-being. (Herbert et al.)

To achieve meaningful change, students should advocate for increased transparency in university spending, request clear reporting on the allocation of OER savings, and support reforms that address the comprehensive cost of higher education. Faculty play a critical role by seeking equitable compensation and recognition for OER contributions, including requesting release time, stipends, or professional development funding. Establishing faculty committees or working groups on OER policy ensures that faculty perspectives are incorporated into decision-making processes. Faculty should also pursue clear guidelines regarding ownership rights and encourage institutions to acknowledge OER work in promotion and tenure evaluations. Collaboration with academic senates and ongoing communication with administration are essential for shaping policies that support both faculty workload and student learning. (Becker et al. 1-20) Leadership from presidents and provosts is essential. Senior administrators can foster transparency by sharing detailed budget information, setting explicit expectations for OER savings, and establishing policies that ensure fair compensation and recognition for faculty. By demonstrating openness and supporting these initiatives, institutional leaders can build trust and show a genuine commitment to affordability and educational excellence.

Belikov and Bodily (2016) found that institutional motivations often diverge from publicly stated values. While OER initiatives are presented as supporting "student success" and "access," internal documents and interviews reveal that actual priorities include cost reduction, positive publicity, and enrollment advantages. Only 23% of institutions with OER mandates monitor student learning outcomes to ensure quality. Although concerns about implementation persist, research suggests that student learning outcomes with OER are generally comparable to or slightly better than those with traditional materials (Hilton, 2020). Continued monitoring is necessary to maintain high academic standards across courses and institutions. To improve transparency and protect educational quality, institutions should require regular monitoring and public reporting on OER impacts. This includes systematic collection and publication of data on student learning outcomes, cost savings, and faculty and student feedback. Ongoing reporting helps ensure OER adoption aligns with institutional goals, supports student success, and maintains academic standards. (Herbert et al.) To ensure educational quality, instructors should follow best practices when selecting or adapting OER materials. Faculty should thoroughly review OER content for accuracy, alignment with course objectives, accessibility, and cultural inclusivity. Comparing multiple OER options, consulting peer or expert reviews, and piloting materials before full adoption help maintain high standards. Many institutions support this process through libraries or teaching and learning centers, offering assistance with locating, evaluating, or adapting OER, as well as workshops, grants, or consultations. Collaborating with department liaisons or librarians can further ensure quality. Incorporating feedback from students and colleagues and regularly updating materials also promotes positive learning outcomes. These practices help instructors select OER resources that align with curriculum goals and meet students' needs. (Orzech et al. 1-20)

The Major OER Repositories

OpenStax, Rice University, Houston, Texas

OpenStax, based at Rice University, is a leading provider of open educational resources (OER) for high-enrollment introductory courses in North America. Established in 2012 with philanthropic support, OpenStax aims to match the quality and rigor of commercial textbooks while offering them free online and at low cost in print.

OpenStax follows a traditional publishing model: subject-matter experts author each title, which undergoes multiple peer reviews and professional editing. Ancillary materials include test banks, instructor manuals, lecture slides, LMS integration, and partnerships with homework platforms. By 2022, over 3.6 million students at 2,500 institutions used OpenStax in 14,000 courses, saving an estimated $1.5 billion in textbook costs (OpenStax, 2022). OpenStax publishes about 60 core textbooks across major disciplines, all aligned with AP and college standards, regularly updated, and meeting WCAG accessibility requirements. This professional approach makes OpenStax a low-risk option for departments prioritizing quality and student preparedness. Institutions benefit from student savings, improved retention, alignment with affordability goals, and progress toward student success and equity. These factors make OpenStax a strategic resource for institutions seeking academic rigor and measurable outcomes.

Faculty may begin by browsing the OpenStax catalog at openstax.org, reviewing full texts and ancillary materials, and selecting titles that align with course requirements. Textbooks are available in multiple formats and can be adopted as-is, customized using tools such as Pressbooks, or integrated into learning management systems via Common Cartridge files or direct links. Most titles are compatible with major LMS platforms, facilitating straightforward adoption and distribution. Formal institutional approval is usually not required.

The BCcampus Open Collection, Victoria, British Columbia, Canada

The BCcampus Open Collection in British Columbia, Canada, is a government-funded leader in open education and one of the most established provincial OER programs globally. Launched in 2012, it aims to lower student costs in the province's public post-secondary system and build faculty capacity.

The collection includes over 350 open textbooks and course packages, with strengths in trades, health, adult basic education, and arts and sciences tailored to the Canadian context. Since 2012, these resources have been adopted in more than 1,000 courses at 40 institutions, saving students about $20 million CAD as of 2021, with continued growth (BCcampus, 2021). The collection features both BCcampus-funded original textbooks and selected, reviewed external resources. BCcampus stands out for its structured quality assurance, offering detailed reviews using standardized rubrics for comprehensiveness, accuracy, relevance, pedagogy, cultural relevance, and accessibility. Faculty reviewers are compensated, and their reviews are published alongside each textbook. BCcampus also supports adaptation through open pedagogy training, Pressbooks hosting, and funding for Indigenous content localization.

The Open Textbook Library, University of Minnesota, Minneapolis

The Open Textbook Library (OTL) at the University of Minnesota is a curated catalog and discovery platform, not a publisher. It lists over 1,100 open textbooks from sources such as OpenStax, BCcampus, Saylor Academy, and independent faculty, organized in a searchable, subject-based database for faculty use. Its main strengths are curation and large-scale peer review.

By 2023, OTL-listed textbooks had been adopted in thousands of courses at hundreds of institutions, although precise usage data are unavailable because OTL does not host the books. Quality assurance comes from a faculty review program, where faculty across North America write structured, criteria-based reviews and receive a small stipend. These reviews and ratings are publicly available, offering peer-to-peer insights and detailed evaluations; however, consistency varies, and overall quality depends on the source. Some titles are highly polished, while others are more basic. The OTL serves mainly as a discovery and vetting tool. Faculty are advised to consult multiple reviews, preview content, check for ancillary materials, and seek input from colleagues before adopting an OER. This process supports informed decision-making.

LibreTexts, University of California, Davis

LibreTexts is the largest and most technologically advanced collaborative project in the OER ecosystem. Originating from the UC Davis ChemWiki in 2008, it now offers over 150,000 pages of open content across 13 libraries, including Chemistry, Mathematics, Biology, Physics, Engineering, Medicine, Humanities, Social Sciences, and Workforce. In 2021, LibreTexts materials were accessed over 1 billion times, used by more than 5 million students monthly, and adopted in over 12,000 courses at 1,500 institutions, making it the highest-traffic OER platform (Delmas et al., 2022).

LibreTexts functions as a "living library" designed for remixing. Faculty can legally and technically adapt, reorder, and add content in real time, combining chapters from multiple sources into a custom textbook with automatic numbering, indexing, and LMS integration. The Chemistry section is particularly extensive and maintained by a large community. This dynamic model enables rapid updates but raises concerns regarding quality control and version consistency, as many authors continually update their content. Faculty are advised to lock or "fork" a stable version for their course to avoid mid-semester changes. Although this process requires some technical skill, it offers exceptional flexibility. For those less comfortable with technology, LibreTexts provides extensive support, including documentation, guides, webinars, and direct assistance, to ensure broad faculty participation.

The American Institute of Mathematics (AIM) Open Textbook Initiative, San Jose, California

The American Institute of Mathematics (AIM) Open Textbook Initiative focuses solely on mathematics textbooks and uses a highly selective, quality-first approach. AIM approves only titles deemed “high quality and open source” after a formal editorial review.

As of 2023, AIM lists about 90 approved textbooks, covering topics from precalculus to advanced undergraduate and early graduate subjects such as abstract algebra, real analysis, topology, and number theory. Each book is rigorously reviewed for mathematical accuracy, clarity, teaching effectiveness, and proper open licensing. Reviews are conducted by mathematicians with relevant teaching expertise, providing a level of quality assurance uncommon in larger repositories. For mathematics departments prioritizing precision and quality, AIM’s list offers a trusted, curated shortlist that reduces the vetting workload.

Other Significant OER Repositories and Platforms

Beyond these five major hubs, the OER ecosystem encompasses several other significant repositories that serve various specialized niches.

MERLOT, founded in 1997 at California State University, offers over 40,000 peer-reviewed resources, including full courses, learning objects, simulations, and assignments, reviewed by discipline-specific editorial boards. OER Commons, created by ISKME, provides more than 100,000 resources and an Open Author tool for building and sharing lessons, with strong coverage of K-12 and community colleges.

MIT OpenCourseWare is notable for providing open access to materials from over 2,400 MIT courses, including syllabi, lecture videos, assignments, and exams. This resource enables faculty to examine how leading institutions structure courses, even when the materials are not intended as standalone textbooks. Additional key resources include OpenLearn from the Open University UK, which offers nearly 1,000 free courses; Saylor Academy’s collection of over 100 fully open, credit-recommended courses with aligned textbooks; and subject-specific platforms such as Project Gutenberg, which provides over 70,000 public domain literary texts, PhET Interactive Simulations from the University of Colorado Boulder for interactive science and math, and Khan Academy for supplemental video and practice across multiple subjects. Collectively, these platforms form a robust, interconnected ecosystem that allows faculty to combine core open textbooks with simulations, primary sources, and practice tools.

OER Adoption Statistics and Trends:

Determining exact adoption rates is challenging because institutions track and report data differently, and there is no single national standard for defining “adoption.” For this analysis, 'OER adoption' means using open educational resources as the main required textbook or course materials for a course section, not as supplementary resources. Some colleges count any course using at least one open resource, while others count only those where OER is the primary required textbook. Tracking methods vary, including by course sections, unique courses, or faculty self-report. The most reliable national estimates come from Seaman and Seaman’s ongoing Bay View Analytics surveys of U.S. faculty, which found that about 8-10% of higher education faculty in the United States used OER as the main required textbook for at least one course in 2021, up from about 3% in 2015-2016. Thus, OER adoption has more than doubled over five to six years, showing significant growth, though it still accounts for a small share of all courses. Adoption rates by discipline vary considerably. For example, in mathematics and statistics, introductory courses like College Algebra and Statistics have adoption rates between 12 and 18 percent, supported by strong OER options like OpenStax and AIM-approved texts. In the social sciences, introductory Psychology courses see adoption rates of 15-20 percent, largely due to widely recognized OER such as OpenStax Psychology and NOBA.

In contrast, humanities and upper-division STEM and professional courses have much lower adoption, typically ranging from 3-10 percent, reflecting gaps in high-quality, comprehensive OER for those subjects. More than 90% of courses still rely primarily on commercial publisher materials, publisher inclusive access programs, or have no required textbook. At this institution, current data indicates that approximately [insert local OER adoption rate]% of course sections use OER as their primary required textbook, which is [above/below/on par with] the national average. Adoption is strongest in [insert local leading disciplines, e.g., mathematics, psychology], consistent with broader national trends, while other areas remain underrepresented. This comparison highlights both progress to date and opportunities for targeted growth where established models and resources exist.

Adoption rates vary widely by institution type, control, and state policy environment, reflecting differences in student need, faculty incentives, and funding. Seaman and Seaman (2021) found that community colleges have consistently higher OER adoption — approximately 12-15% of courses — than four-year public universities (6-9%) and four-year private institutions (4-6%). Public institutions use OER at roughly twice the rate of private institutions (10% compared to 5%). This pattern is logical: community colleges serve a higher proportion of low-income, Pell-eligible students for whom textbook costs are a direct barrier to persistence, and they have more centralized support for OER through libraries and teaching centers. States with dedicated OER funding programs, legislative mandates, and grant programs — like California with its OER Council and Zero Textbook Cost degree program, Georgia with Affordable Learning Georgia, Washington with OpenWA, and Virginia with VIVA Course Redesign Grants — have adoption rates two to three times higher than states without such support, because funding provides release time, stipends, and instructional design help. Large introductory courses also use OER far more (15-25% of sections) than upper-division or specialized courses (3-7%) because high-quality, professionally produced alternatives like OpenStax exist and because savings scale to thousands of students.

Lower-division courses use OER much more than upper-division and graduate courses, a pattern driven almost entirely by availability of suitable materials. Exact numbers depend on the institution and subject, but research shows a clear tiering. Introductory general education courses such as English Composition, College Algebra, Introduction to Psychology, Principles of Economics, General Biology, and General Chemistry have OER adoption rates of 15-30% at schools with active OER programs and up to 40% at some Zero Textbook Cost pathways. These are high-enrollment courses where excellent OER exists and where faculty teach common learning outcomes. Upper-division major courses have adoption rates of 5-10%, often limited to a few faculty champions who create their own materials. Specialized, advanced, or graduate courses have very low adoption — 2-5% or less — mainly because there are not enough high-quality, current, and specialized OER materials for niche topics like advanced organic synthesis, specialized nursing specialties, or graduate-level econometrics, and because faculty in those courses often rely on primary literature or highly specialized commercial texts (Seaman & Seaman, 2021; Croteau, 2017).

State-level initiatives dramatically increase adoption and demonstrate what coordinated investment can achieve. Florida Virtual Campus's 2021 survey of the Florida College System and State University System reported that Florida's OER initiatives, including open textbook grants and the Florida Open Academy, resulted in over 6,000 course sections adopting OER materials across Florida's 40 public colleges and universities in a single year, representing approximately 12-15% of high-enrollment general education courses and saving students an estimated $20 million annually. The state achieved this through centralized cataloging in FLVC and modest faculty grants. Similarly, Affordable Learning Georgia (Croteau, 2017) documented over 1,000 course sections at University System of Georgia institutions adopting OER by 2017, with a concentration in lower-division general education courses at both research universities and access institutions. Georgia's model is notable because it tied OER grants to continuous improvement grants and tracked actual student savings, documenting over $50 million in cumulative savings by 2020, showing that sustained, centralized funding produces measurable returns.

California's ambitious OER initiatives show both the promise and the limits of large-scale investment without mandates. California's large OER programs have had mixed results despite significant investment. The California OER Council, the Open Educational Resources Initiative, and the California Community Colleges Chancellor's Office have spent over $30 million combined on OER development, adoption grants, and the creation of Zero Textbook Cost (ZTC) degree pathways. Ozdemir and Hendricks (2017) found that about 10-15% of courses at California community colleges use OER in at least one section, with higher rates of 20-30% in high-enrollment, high-cost courses like English Composition, Elementary Statistics, College Algebra, and Introduction to Psychology, where OpenStax and other quality options exist. Other subjects and upper-division courses have much lower rates — 5% or less — because suitable materials are scarce. In 2020, the California State University system reported OER use in over 2,000 course sections serving more than 76,000 students each year, but this represents less than 5% of all course sections across its 23 campuses. The lesson from California is that even with millions in funding, adoption remains concentrated and voluntary. Without changes to tenure incentives, bookstore contracts, and inclusive access deals, it does not automatically scale to all disciplines.

Adoption patterns also vary dramatically by discipline, reflecting the maturity of the open corpus in each field. Seaman and Seaman (2021) and related discipline-specific studies found that mathematics and statistics show relatively high OER adoption (12-18 percent of introductory courses) due to the ready availability of high-quality, peer-reviewed OER such as OpenStax College Algebra, Statistics, and Calculus, and the AIM-approved textbook list, which provides quality assurance. Science disciplines show moderate adoption (10-15 percent in introductory Biology, Chemistry, and Physics), primarily driven by OpenStax materials, PhET simulations, and LibreTexts, where adoption is often department-wide. Social sciences vary widely: Psychology shows 15-20 percent adoption thanks to OpenStax Psychology and NOBA; Sociology shows 10-15 percent adoption; Economics shows 8-12 percent adoption, with OpenStax Principles covering the intro sequence; and Political Science shows only 5-8 percent adoption, where American government OER is newer. Humanities show low adoption (5-10 percent overall), except for some literature courses that use public-domain texts from Project Gutenberg, because faculty often prefer specific scholarly editions and anthologies. Professional programs show very low adoption (3-5 percent in nursing, business, engineering, and education), due to lack of current, specialized OER, rapid changes in industry standards, and strict accreditation requirements that specify certain commercial texts or software platforms.

Globally, however, OER adoption rates and trends vary significantly. While North America remains a leader in OER development, curation, and formal adoption in higher education, some countries—such as South Africa, the United Kingdom, and the Netherlands—have implemented national OER policies and seen strong growth in usage, especially at the secondary level. In many regions including Latin America, Eastern Europe, and Asia, initiatives to promote OER are growing, but adoption levels in higher education still lag behind North American rates. International comparisons highlight that local policy environments, funding models, and language availability greatly influence both faculty awareness and institutional uptake.

OER adoption is growing quickly in percentage terms, but it started from a very low baseline and may be approaching a plateau without new structural support. Seaman and Seaman (2021) report that OER use as a primary textbook increased by 150-200% from 2016 to 2021, rising from about 3-4% to 8-10% of courses nationally. This is a significant percentage increase and represents a cultural shift in faculty awareness. However, it still reaches only a minority of students, and growth has been slowing since 2019. Researchers warn that without major new funding, sustainable faculty incentive models, and institutional requirements such as ZTC degree pathways or automatic consideration of OER during textbook adoption, OER adoption may plateau at 15-20% of courses, mostly concentrated in lower-division general education, rather than becoming the norm across the curriculum. The remaining 80% will continue to use commercial materials unless inclusive access contracts and publisher platforms become the default cost-saving mechanism instead.

To address this plateau and accelerate progress, two actionable steps are recommended. First, allocate dedicated funding for faculty stipends, release time, or department-level OER initiatives to directly address resource and incentive barriers to adoption. Second, establish a formal policy requiring regular review of OER alternatives in all textbook selection cycles for high-enrollment courses, ensuring that consideration of OER is integrated into standard institutional practice. These actions can generate meaningful momentum and expand OER use beyond current limits.

The impact of the COVID-19 pandemic on OER adoption remains unclear and somewhat counterintuitive. The effect of the COVID-19 pandemic on OER adoption is still uncertain and is being studied. Initially, many advocates hypothesized that the sudden move to remote learning in Spring 2020 would dramatically accelerate OER adoption due to concerns about students' physical access to campus bookstores and shipping delays. They heightened financial hardship, as well as the ease of linking to free digital materials in an LMS. However, Seaman and Seaman (2021) found mixed and modest results. Some institutions with existing OER infrastructure did see increased interest and new adoptions during 2020-2021, as faculty sought materials they could legally post and adapt. But many faculty, overwhelmed by emergency remote teaching, the need to learn new technology, and caregiving responsibilities, chose to keep using familiar commercial materials rather than switching to new ones, especially when publishers temporarily offered free e-book access during the pandemic. In some cases, publisher-inclusive access programs that automatically bill students actually expanded during the pandemic. The long-term effects of the pandemic on OER adoption remain unclear and will likely depend on whether institutions retain the focus on affordability and digital equity that emerged in 2020.

Institutional Mandates and Incentives:

A growing number of institutions have implemented OER mandates, strong incentives, or full degree pathways designed to eliminate textbook costs. Ozdemir and Hendricks (2017) documented that, as of 2017, at least 40 U.S. institutions — primarily community colleges — had established "OER degrees" or "Z-degrees" (zero-cost textbook degrees or Zero Textbook Cost pathways), in which all or nearly all required courses for a specific credential use OER, library-licensed, or other no-cost materials. These were pioneering efforts, often launched at colleges like Tidewater Community College in Virginia, which created the first Z-Degree in Business Administration in 2013. By 2021, this number had grown to over 100 institutions and 400+ distinct Z-degree pathways nationwide, according to SPARC's tracking (SPARC, 2021), with strong concentrations in California, Virginia, Texas, and Washington, where state funding supported their development.

However, the scale and scope of these programs remain limited. These programs typically serve small numbers of students relative to total institutional enrollment — often 50 to 300 students per pathway — and concentrate in specific majors where OER availability is strongest and most mature, such as general studies, liberal arts, general education transfer, business administration, and social sciences like psychology and sociology. They are far less common in high-credit STEM, nursing, and technical majors where commercial labs, codes, and specialized textbooks are difficult to replace. For many institutions, the Z-degree is also more of a ZTC (Zero Textbook Cost) degree than a pure OER degree, because faculty are allowed to use library e-books, faculty-created materials, and other no-cost resources to fill gaps where true openly licensed OER does not exist. While powerful symbols of commitment to affordability, Z-degrees often require intensive, ongoing faculty collaboration, centralized curation, and dedicated advising to keep pathways truly zero-cost as faculty and editions turn over.

State-level mandates and funding create significant top-down pressure for adoption, fundamentally changing the incentive structure for institutions and faculty. At least 15 states have enacted legislation or funding initiatives that promote, incentivize, or require OER adoption in public institutions, with the number growing to over 20 states by 2023 (SPARC, 2021). These policies exist on a spectrum of strength and approach. The weakest are informational: requirements that institutions track and report OER usage and mark courses as low-cost or zero-cost in the course schedule and registration system so students can find them, as implemented in Florida, Texas, California, and Oregon. The mid-range are financial incentives: competitive grants for OER development, adaptation, or adoption, as seen in California's $115 million ZTC program, Georgia's Affordable Learning Georgia, Washington's OpenWA grants, Virginia's VIVA Course Grants, Ohio's OhioLINK, and New York's state-funded OER initiative. The strongest are quasi-mandates: requirements that OER be formally considered for high-enrollment general education courses before a commercial text can be adopted, or that institutions develop plans to reduce costs, as seen in Oregon's HB 2729 and Maryland's initiatives.

This policy environment has undeniably increased adoption, but it has also created new tensions. Belikov and Bodily (2016) found in their survey of over 200 faculty members that faculty in states with such policies feel increased pressure, both explicit pressure from the administration and implicit pressure from reporting requirements, to adopt OER regardless of quality or pedagogical fit. Faculty reported concerns that textbook selection decisions, traditionally protected under academic freedom, were being influenced by cost-savings targets set at the state or system level. When adoption is tied to departmental savings goals, bookstore revenue models, or performance funding, faculty may feel compelled to adopt materials that are "good enough" rather than the best fit for their learning objectives, especially in upper-division courses. This highlights the critical need for policies that pair incentives and expectations with robust quality review, faculty autonomy in selection, and sustained support for adaptation, rather than simply mandating cost reduction.

To address these challenges and promote sustainable, collaborative adoption, institutions should engage faculty through transparent decision-making processes, faculty-led OER committees, and recognition of OER-related work in tenure and promotion guidelines. Early involvement of faculty in the selection and adaptation process helps build trust and ensures that educational quality remains central. Institutions should also provide optional professional development, constructive peer review, and structured pilot programs that allow for feedback and iteration prior to broader adoption. Balancing cost-saving directives with ongoing faculty input and rigorous quality assurance reassures institutional stakeholders and trustees that OER adoption will enhance, rather than compromise, educational outcomes.

What Statistics Reveal:

Despite growth and investment, OER remains a minority practice affecting 8-10% of courses overall nationally, and its distribution is highly uneven rather than broad-based. The data show that adoption is concentrated heavily in five overlapping niches rather than spread evenly across the curriculum: First, lower-division general education courses — the high-enrollment English Composition, College Algebra, Psychology, Sociology, and introductory science courses that form the general education core — where adoption reaches 15-30% at institutions with active programs. Second, introductory STEM and social sciences specifically where mature, peer-reviewed, professionally produced quality OER exists, such as OpenStax Biology, Chemistry, Physics, Psychology, and Economics, with adoption of 10-20% in those disciplines. Third, community colleges rather than four-year institutions, with 12-15% adoption versus 6-9% at four-year publics and 4-6% at private institutions, because community colleges serve the most price-sensitive students and have stronger centralized support from libraries. Fourth, institutions located in states with dedicated OER funding and mandates or transparency laws — California, Georgia, Washington, Virginia, Florida, and Texas — where adoption reaches 15-20% compared to 5-7% in states without such support. And fifth, specific institutions with aggressive, decade-long OER initiatives and dedicated staff, such as Tidewater Community College, the University of Georgia, and some California State University campuses, some of which report 25-40% of course sections using OER or other zero-cost materials when broadly defined. Outside these concentrated pockets, adoption falls off sharply.

The vast majority of courses — 90%+ overall, and 95%+ of upper-division, graduate, and specialized professional courses — still use commercial textbooks, publisher inclusive access programs, or other non-OER materials. This persistent dominance suggests three non-mutually exclusive explanations that research has identified. The first is a supply problem: quality, comprehensive, and current OER doesn't exist for most courses, especially specialized upper-division courses like Advanced

Pathophysiology, Quantum Mechanics II, Advanced Corporate Tax, or niche humanities seminars where the market is too small to attract OER authorship. The second is a perceived quality and fit problem: that faculty have reviewed available OER and judge it to be inadequate for their specific courses — lacking depth, problem sets, currency, cultural relevance, or alignment with their pedagogical approach — even when something exists. The third is an incentive and structural problem: that insufficient incentives, recognition, time, and support exist to overcome faculty concerns and the substantial switching costs of redesigning a course around new materials, especially when tenure and promotion systems do not reward OER work.

The fact that OER is so heavily concentrated in lower-division general education courses where commercial textbooks are most expensive, most standardized, and most visible to legislators and the media suggests strategic targeting by institutions — a rational triage approach. Institutions have focused first on addressing the most visible, politically sensitive textbook costs where savings are largest, and materials are available, rather than attempting a comprehensive OER transformation across all levels of the curriculum. It is an access strategy that maximizes reported student savings with minimal disruption, while leaving the majority of the curriculum untouched.

If adoption continues at current historical growth rates — roughly doubling every 5-6 years — OER might reach 15-20% of courses by 2025-2027 and 25-30% by 2030, but that trajectory is far from guaranteed. This projection assumes three conditions that are not currently secured: continued and expanded state and federal funding for OER creation and adoption grants and for Zero Textbook Cost degree pathway staffing; the sustained development of high-quality, accessible, and ancillary-rich OER for currently underserved subjects and upper-division courses, which requires far more specialized author teams and ongoing maintenance; and growing faculty acceptance of OER quality and pedagogical fit, which requires evidence of efficacy and respect for academic freedom.

Without directly addressing quality concerns documented in research (Bliss et al., 2013; Belikov & Bodily, 2016) (including concerns about comprehensiveness, accuracy, lack of supplementary materials like test banks and adaptive homework, and the increased faculty workload required to vet and adapt OER), adoption is likely to plateau well below majority practice. Research on innovation diffusion suggests a chasm between early adopters motivated by ideology and the early majority who require proven quality, ease of use, and institutional reward. Many institutions are now encountering that chasm, where the next 10 percent of adoption is significantly harder than the first 10 percent because it requires moving beyond enthusiastic champions to faculty who are skeptical, time-constrained, or teaching specialized content. Without addressing these structural and quality barriers, OER will remain an important but minority alternative rather than the default.

In response to these persistent barriers, several emerging strategies are targeting upper-division and specialized courses. New grant-funded pilot projects at universities such as UC Davis and the University of Minnesota are supporting cross-institutional faculty teams to develop advanced OER in subjects like environmental engineering and molecular biology, often coordinated with disciplinary associations or subject consortia. Collaborative models, including partnerships between community colleges and research universities, are pooling expertise and sharing development costs for upper-division textbooks and course materials. Some states, like Texas and New York, are providing dedicated funding to create specialized OER, with a focus on supporting advanced courses in STEM, nursing, and business where gaps have been identified through faculty needs assessments. There is also growing interest in adapting open-source software and interactive platforms, such as Jupyter Notebooks and LabArchives, to create interactive OER for lab-based and data-intensive upper-division courses. While these approaches are still at a pilot stage, they represent promising innovations aimed at closing the current coverage gap in advanced and specialized curricula.

To overcome these obstacles, institutions should implement collaborative textbook vetting processes in which faculty committees review and select OER collectively, sharing the workload and building consensus on quality. Providing structured institutional support such as stipends, course release time, or dedicated OER librarians can help compensate for the time faculty invest in evaluating and adapting materials. Establishing shared repositories for adaptation notes, revisions, and supplementary materials (such as test banks and homework sets) enables instructors to benefit from each other's work, reducing redundant effort. Integrating standardized OER quality frameworks into review processes can further strengthen adoption efforts and ensure consistency. Widely used rubrics such as the BCcampus Open Textbook Review Criteria and the Achieve OER Rubrics offer comprehensive tools for evaluating aspects such as accuracy, comprehensiveness, accessibility, and pedagogical effectiveness. These frameworks support faculty in making informed selections and facilitate comparison of OER across sources. Finally, professional development sessions focused on best practices for adopting and customizing OER can empower more faculty to participate, ensuring quality while minimizing individual workload burdens.

The Quality Problem: Research Evidence of OER Limitations and Gaps

OER Quality Is Highly Variable and Often Inadequate:

Research on OER quality reveals significant concerns. Delimont et al. (2016) surveyed 108 educators and found that perceived barriers to OER development included: lack of time (91%), lack of institutional support (58%), lack of technical expertise (53%), concerns about quality control (47%), and lack of funding (45%). These barriers suggest that much OER is created under less-than-ideal conditions. Delimont et al. (2016) surveyed 108 educators who had been involved in OER development or adaptation at multiple institutions and found that perceived barriers to creating high-quality OER were systemic rather than attitudinal. The top barrier was lack of time, cited by 91% of respondents — faculty are typically asked to create OER on top of existing teaching, research, and service loads without release time. The second was lack of institutional support at 58%, including lack of instructional designers, copyright librarians, accessibility reviewers, and graphic designers who are standard on commercial publisher teams. Lack of technical expertise was cited by 53%, as many faculty lack training in Pressbooks, LaTeX, accessibility remediation, or open licensing. Concerns about quality control were cited by 47%, reflecting worry that their work would not be peer-reviewed or maintained. And lack of funding was cited by 45%, with most grants covering only a small stipend. These barriers suggest that much OER — particularly OER created outside of well-funded projects like OpenStax — is created under less-than-ideal, volunteer conditions by isolated individuals rather than by supported, interdisciplinary teams, which inevitably affects consistency, production value, and long-term sustainability.

Quality assessments of OER reveal substantial gaps. Bliss et al. (2013) evaluated the quality of OER using the LORI (Learning Object Review Instrument) framework and found significant variability in quality, with many OER scoring poorly on content quality, learning goal alignment, feedback and adaptation, and motivation. The study noted that "the quality of OER is highly variable" and recommended careful evaluation before adoption. Bliss et al. (2013) evaluated the quality of a random sample of open textbooks and open courseware using the LORI (Learning Object Review Instrument) framework, a nine-dimension instrument measuring content quality, learning goal alignment, feedback and adaptation, motivation, presentation design, interaction usability, accessibility, reusability, and standards compliance. They found significant variability in quality, with many OER scoring well on reusability and accessibility but scoring poorly on the most instructionally critical dimensions: content quality (accuracy, balance, and comprehensiveness), learning goal alignment (whether assessments and activities actually matched stated objectives), feedback and adaptation (whether materials provided formative feedback and pathways for different learners), and motivation (whether materials engaged learners). The study noted that "the quality of OER is highly variable" and that a high average rating masked a bimodal distribution where a few polished OER were excellent and many were poor, leading the authors to recommend careful, course-level evaluation before adoption rather than assuming openness equals effectiveness. This finding has been replicated in later studies using Achieve OERR and other rubrics.

Faculty report concerns about the quality and completeness of OER. Allen and Seaman (2014) found that only 11.3% of teaching faculty were aware of and using OER, and Belikov and Bodily (2016) found that faculty concerns about OER quality were the primary barrier to adoption, with faculty reporting that OER often lacked comprehensiveness, had gaps in content coverage, and required significant adaptation and supplementation. Allen and Seaman (2014), in one of the earliest national surveys, found that only 11.3% of teaching faculty were both aware of and using OER, but among those who were aware, quality concerns were prominent. Building on that, Belikov and Bodily (2016) surveyed over 200 faculty who had considered OER and found that faculty concerns about OER quality were the primary and most frequently cited barrier to adoption, outranking concerns about time or technical skills. Faculty reported in open-ended responses that OER often lacked comprehensiveness compared to commercial texts they were replacing, had significant gaps in content coverage for their specific course objectives, was outdated or lacked currency on rapidly changing topics, and required significant adaptation, supplementation, and curation to be usable. Many described a "Swiss cheese" problem — the core text might be solid, but chapters were missing, the index was weak, or ancillaries like test banks did not exist. This creates a hidden workload transfer: the cost savings to students are subsidized by uncompensated faculty labor to make the materials viable, a concern particularly acute for contingent faculty and those teaching heavy loads.

In STEM fields, particularly, OER problem sets are often inadequate. Winitzky-Stephens and Pickavance (2017) found that faculty in mathematics and sciences reported that available OER lacked sufficient practice problems, worked examples, and variety of problems—essential components for developing quantitative and analytical skills. Faculty reported spending 10-20 additional hours per course creating supplementary problems to address OER gaps. Winitzky-Stephens and Pickavance (2017) interviewed and surveyed faculty in mathematics, statistics, chemistry, and physics who had adopted or piloted OER and found that available OER frequently lacked sufficient quantity, variety, and cognitive complexity of practice problems, step-by-step worked examples, and conceptual questions — essential components for developing quantitative, analytical, and problem-solving skills where learning happens through doing. Unlike humanities texts where reading is central, STEM learning depends on deliberate practice with feedback. Faculty reported that OER problem sets often had too few problems per learning objective, lacked problems at varying difficulty levels from procedural to conceptual, lacked real-world application problems, and lacked fully worked solutions that model expert thinking. To address these gaps and maintain rigor, faculty reported spending an additional 10 to 20 hours per course, per semester, creating supplementary problems, sourcing problems from other open sources, or building their own homework sets in WeBWorK or MyOpenMath. This not only increases workload but also raises equity concerns, because students in sections where faculty have time to supplement receive a richer practice environment than those where faculty do not, creating inconsistent learning experiences.

Peer Review and Vetting Gaps:

Traditional textbooks from major publishers undergo an extensive, capital-intensive, multi-year peer review and editorial process that is designed to minimize errors and maximize classroom usability. Pawlyshyn et al. (2013) document that commercial textbook development for a major introductory title is more akin to product development than individual authorship. The process involves multiple rounds of expert review at the proposal, outline, chapter draft, and final manuscript stages, including reviews by 50 to 100+ faculty specialists for accuracy, coverage, and pedagogical approach. It involves dedicated developmental editing to ensure consistent voice, appropriate reading level, and alignment between learning objectives, exposition, examples, and assessments. It includes professional fact-checking, copyediting, indexing, and accessibility compliance, plus student and instructor pilot testing and class-testing where draft chapters are used in real courses to collect learning data and feedback. The full cycle typically takes 2 to 4 years of development with coordinated teams of 10 to 30 experts including authors, developmental editors, art directors, photo researchers, assessment specialists, and digital learning designers. Publishers invest $500,000 to $1,000,000 or more in major textbook development, and often far more for best-selling titles with integrated platforms, precisely because their business model depends on perceived quality, adoption by departments, and longevity. While this system is not without flaws — including incentives to create cosmetic new editions — it does provide layers of quality control that are difficult to replicate with volunteer labor.

OER, by contrast, lacks a comparable systematic, standardized, and funded vetting infrastructure across most of the ecosystem. While some repositories like MERLOT and OpenStax do have formal review processes — MERLOT uses discipline editorial boards and OpenStax mirrors commercial peer review — these represent the exception rather than the rule. Much OER is self-published directly by faculty authors to institutional repositories, personal websites, or Pressbooks instances without any external review beyond perhaps a single colleague looking it over. Jung et al. (2017) audited content in major OER repositories and found that only 23% of OER in major repositories had undergone any form of formal, documented peer review prior to publication, compared to nearly 100% of commercial textbooks from major publishers. Clements and Pawlowski (2012) identified quality assurance as one of the most significant and persistent challenges in the OER movement, noting that "lack of quality assurance mechanisms is a major impediment to OER adoption" because faculty cannot easily signal trustworthiness to peers. The result is a highly variable landscape where excellent, peer-reviewed OER sits alongside incomplete drafts, outdated notes, and well-intentioned but inaccurate materials, with little labeling to distinguish them, forcing individual faculty to become the de facto quality control system.

This places a substantial and often invisible evaluation burden on individual faculty, who lack the time and sometimes the specialized expertise to thoroughly vet OER quality at the level a publisher team would. Belikov and Bodily (2016) found that faculty who seriously considered OER adoption reported needing 20 to 40 hours to adequately evaluate a single OER textbook before adoption — reading the entire text, checking accuracy against current knowledge, mapping it to their learning objectives, evaluating problem sets, checking figures and accessibility, and reviewing ancillaries — time they often don't have during a busy semester, especially contingent faculty teaching multiple sections. That evaluation burden is particularly challenging outside their subspecialty areas; a biologist teaching general biology may be expert in cell biology but less equipped to vet the accuracy of the ecology or genetics chapters, whereas commercial publisher review assigns those chapters to different specialists. Faculty also report difficulty evaluating technical aspects like accessibility compliance with screen readers, copyright clearance of images, and alignment to accreditation standards. When this vetting time is uncompensated and unrecognized in tenure and promotion, rational faculty may decide that the switching costs outweigh the benefits, especially if they have already invested years in refining lectures and assessments around a commercial text they trust.

Currency and Updating Problems:

Keeping content current is essential for learning, not a cosmetic preference, particularly in rapidly evolving fields where accuracy directly impacts student competence and safety. Research shows that outdated materials measurably harm learning outcomes, motivation, and trust, especially in STEM, technology, healthcare, business, and social sciences where knowledge changes quickly. In biology, for example, outdated genetics content may lack CRISPR; in chemistry, outdated safety protocols may be dangerous; in computer science, a 5-year-old text may teach deprecated languages; in nursing and allied health, outdated guidelines can lead to unsafe practice; in economics and political science, outdated data can fundamentally misrepresent reality. Fischer et al. (2015) found that students who used outdated or inaccurate versions of open textbooks performed significantly worse on learning assessments than students using current versions, even when controlling for prior knowledge, and reported lower confidence in the materials. When students detect outdated information — a broken link, an old statistic, a reference to a 2012 election as "upcoming" — their trust in the entire resource collapses, and they disengage. Currency signals care and credibility.

OER updating is inconsistent, decentralized, and often abandoned, creating a sustainability crisis often referred to as the "orphan OER" problem. Pitt (2015) tracked the lifecycle of a sample of OER funded through major UK and US grants and found that many OER materials become "orphaned" when their original authors leave institutions, change teaching assignments, lose grant funding, or simply stop maintaining them because there is no institutional expectation or reward for maintenance. Unlike commercial publishers, whose entire business model depends on releasing new editions every 2-3 years to protect revenue streams and maintain adoptions — ensuring ongoing investment in revisions — OER lacks sustainable, funded maintenance structures. Pitt found that 40-60% of OER in major repositories like MERLOT, OER Commons, and early Connexions collections were more than 5 years old, had not been revised since initial publication, and lacked errata, updates, or versioning information. Many had broken links, outdated references, and uncorrected errors reported by users years earlier. Because OER repositories are often archives rather than active publishers, they rarely have the budget to commission updates, and because Creative Commons licenses allow anyone to update but require no one to do so, responsibility falls to a commons that is under-resourced.

Faculty report outdated OER as a significant and practical concern that directly affects adoption decisions. Belikov and Bodily (2016) found that 38% of faculty who had considered OER cited outdated, inaccurate, or not-current-enough content as a primary barrier to adoption — ranking just behind concerns about overall comprehensiveness and lack of ancillaries. This concern was especially pronounced among faculty in health sciences, technology, and current events-heavy social sciences where currency is non-negotiable for accreditation or professional preparation. Delimont et al. (2016) found that even faculty creators themselves often lacked time, funding, or professional incentive to maintain OER after initial creation. Most OER development grants provide one-time funding for creation but no ongoing funding for maintenance, versioning, or community management. After the grant ends, updating becomes volunteer service work that is largely invisible in promotion and tenure files. Faculty reported feeling guilty about abandoning their OER but unable to justify spending 40-60 hours revising a free textbook when that same time could be spent on research, teaching, or service that is rewarded. Without sustainable maintenance models — such as departmental ownership, paid community editors like LibreTexts employs, or institutional commitments to maintain flagship OER — currency will remain a structural weakness of the OER ecosystem.

Supplementary Materials Gap:

Commercial textbooks are not standalone books; they are comprehensive teaching and learning systems that include extensive ancillary materials designed to reduce faculty workload and increase student engagement. Pawlyshyn et al. (2013) documented that a typical commercial textbook package for a large introductory course from a major publisher like Pearson, McGraw-Hill, or Cengage includes far more than the text itself: detailed instructor manuals with course outlines and teaching notes, vetted test banks with 500 to 2,000+ questions organized by learning objective and Bloom's taxonomy level, pre-built PowerPoint slides for every chapter (often 30-50 slides per chapter) with embedded figures, full solutions manuals with step-by-step answers, student study guides and practice exams, integrated online homework systems like MyLab, Connect, or MindTap with adaptive quizzing, algorithmic problem generation, and automatic grading, and multimedia resources such as videos, animations, virtual labs, and case studies — representing millions of dollars in development investment per title. These materials are produced by dedicated teams of assessment specialists, media producers, and learning designers. For many faculty, especially new faculty, contingent faculty teaching multiple sections, and faculty teaching outside their primary research area, these ancillaries are essential infrastructure that enables them to focus on instruction rather than building everything from scratch.

Most OER lacks these materials entirely or has only minimal, uneven versions that do not provide equivalent support. Bliss et al. (2013), using the LORI evaluation framework, found that only 15% of OER in their sample had comprehensive instructor resources that were aligned, complete, and ready to use, and fewer than 10% had any form of test bank or online homework system that could be imported into an LMS. Fewer still had adaptive features. While flagship OER like OpenStax has invested heavily in creating test banks, slides, and partnerships with low-cost homework systems like OpenStax Tutor, Learning Pod, or LibreTexts ADAPT, the vast majority of OER in repositories like OER Commons or the Open Textbook Library consists of a PDF or Pressbook text alone. This creates two significant problems. First, it creates significant additional workload for faculty who must create their own slides, write their own exams, and build their own homework sets. Second, it reduces teaching quality and consistency when individual faculty, without assessment expertise, create ad-hoc materials that may not align with learning objectives, contain errors, or lack variety. The absence of auto-graded homework is particularly consequential in large courses, where immediate feedback is critical but manual grading is impossible.

Faculty consistently report this gap in supplementary materials as a major, practical barrier to OER adoption that outweighs philosophical support for openness. Jung et al. (2017) surveyed faculty across multiple institutions who had considered OER and found that 67% of faculty cited lack of high-quality ancillary materials — especially test banks, slides, and homework systems — as a primary barrier to adoption, ranking it above concerns about content quality. This was particularly true for early-career faculty and for faculty teaching large introductory sections where automated assessment is essential for manageability. Winitzky-Stephens and Pickavance (2017) found in their STEM-focused study that faculty who did adopt OER spent an average of 15 to 20 hours per course creating supplementary materials in the first semester of adoption, and 8 to 10 hours in subsequent semesters maintaining them — time that directly competes with student interaction, mentoring, office hours, research, and course improvement. Faculty described staying up late building PowerPoints because the OER had none, writing exam questions from scratch because no test bank existed, and manually grading homework because no adaptive system was available. This hidden labor is rarely recognized in workload calculations and is a primary reason why some faculty pilot OER and then return to commercial materials, not because they disagree with the mission of affordability, but because the total cost of ownership in terms of time is unsustainable without institutional support.

Faculty Perspectives: Benefits, Burdens, and Mixed Feelings About OER

Why Some Faculty Support OER:

Some faculty are genuinely committed to educational access and equity, viewing OER as a moral imperative rather than just a cost-saving tactic. Kimmons (2015) conducted in-depth interviews with faculty who voluntarily adopted OER without mandates and found that their motivations were deeply values-driven, not transactional. They cited four recurring themes: a philosophical commitment to open education and the principle that knowledge should be a public good, not a commodity locked behind paywalls; a direct, personal desire to reduce financial barriers for students, particularly low-income, first-generation, and community college students they saw skipping meals or dropping out due to textbook costs; deep frustration with commercial publishers' practices — including 300% price increases over 20 years, bundling of expensive access codes, and aggressive tactics to eliminate the used book market — which they saw as exploitation and profiteering off captive students; and a broader commitment to educational justice and equity, where removing cost is a first step toward culturally responsive pedagogy. These faculty often invest enormous amounts of unpaid time — 100+ hours — creating high-quality, openly licensed OER because they believe in the mission, even when their institutions do not reward it. They are the early adopters and moral leaders of the movement.

Some faculty receive tangible benefits or recognition for OER work, though these incentives are uneven and often symbolic. Jung et al. (2017) documented the types of institutional support that do exist: small stipends or course releases for OER creation and adoption, typically ranging from $500 to $5,000 in one-time payments or a single course release for a semester, which is modest compared to the time required; competitive grant funding for larger OER development projects, often from state systems or library-led initiatives; professional recognition and awards such as OER Champion certificates or teaching awards, though these rarely count toward tenure and promotion in research-intensive institutions; conference travel funding to present on OER work at Open Education conferences; and increased visibility and reputation within the national open education community, which can lead to collaborations and consulting. However, these benefits are highly inconsistent across institutions, often temporary and grant-dependent, and frequently inadequate compensation for the time invested. A $1,000 stipend for 60 hours of work equates to less than $17 per hour, and a course release one semester does not cover the ongoing maintenance required for years.

Faculty gain pedagogical autonomy and the ability to customize materials in ways that commercial textbooks legally and technically prohibit. Rather than being constrained by the fixed structure, copyright, and one-size-fits-all approach of a commercial textbook, OER allows faculty to legally and practically: remix and adapt content to fit their specific learning objectives, course sequence, and student needs — reordering chapters, rewriting examples, and cutting irrelevant material; incorporate diverse voices, local case studies, and current events easily and immediately, rather than waiting for a publisher's next edition; create culturally responsive materials that reflect the identities, languages, and communities of their students, which is especially important for Hispanic-Serving Institutions, HBCUs, and Tribal Colleges; update content immediately when new developments occur, such as a Supreme Court decision, a new scientific discovery, or a public health crisis; and retain ownership of their intellectual work rather than signing away rights to publishers and then having to buy back their own content (Petrides et al., 2011). For faculty who see teaching as creative intellectual work, this freedom is profoundly motivating and leads to more engaging, relevant courses.

But Most Faculty Feel Pressured, Burdened, and Concerned:

The majority of faculty, especially those who did not voluntarily seek out OER, report feeling pressured or coerced into OER adoption against their professional judgment. Belikov and Bodily (2016) surveyed over 200 faculty at institutions with formal OER policies, Z-degree initiatives, or strong administrative encouragement and found striking levels of perceived pressure: 63% of faculty at institutions with OER policies felt pressure to adopt OER regardless of quality or pedagogical fit for their specific course; 71% reported inadequate institutional support for OER development and implementation to meet the expectations being placed on them; 58% said OER adoption was "strongly encouraged" in ways that felt like soft mandates — such as departmental savings goals, public tracking dashboards, or direct requests from deans — where saying no felt risky; 47% expressed concerns that OER quality was insufficient for their courses and that they would not have selected it on quality alone; and 82% said they would not have chosen OER without institutional pressure. In qualitative comments, faculty described feeling that their professional autonomy, disciplinary expertise, and years of experience refining their courses were being disregarded in favor of cost metrics, and that questioning OER was framed as not caring about students, which they found deeply demoralizing.

Faculty express deep frustration that OER mandates effectively shift costs from institutions and publishers to their own unpaid labor. Delimont et al. (2016) documented through time diaries that faculty creating or substantially adapting OER spend an additional 20 to 60 hours per course in the first semester, and 10 to 20 hours per subsequent semester for updates — time that is: largely uncompensated, with no extra pay and course releases that are typically inadequate and one-time; invisible to students and administrators who see only "free materials" and not the labor behind them; taken directly from research time, student advising and mentoring, course preparation in other areas, and family time; and systematically undervalued in most institutions, not counting toward tenure, promotion, or post-tenure review in most cases. Petrides et al. (2011) calculated that when valued at standard academic hourly rates for instructional design and subject expertise, this represents $15,000 to $30,000 in faculty labor per course in the first year alone. Faculty resent being told to absorb these costs so institutions can claim large student savings numbers in press releases and legislative reports while simultaneously raising tuition, cutting support staff, and relying increasingly on low-paid contingent labor. They argue true affordability should be funded, not volunteer-subsidized.

Faculty worry, based on their classroom experience, about educational quality and student learning outcomes when forced to use inadequate OER. Surveys consistently show faculty concerns that are pedagogical, not ideological: that OER materials are often less comprehensive and less rigorous than commercial alternatives, with thinner explanations, fewer worked examples, and less robust scaffolding (Bliss et al., 2013); that students are not learning as deeply or retaining material as well when using fragmented OER that lacks coherence, particularly in cumulative subjects where gaps compound (Croteau, 2017); that significant gaps in OER coverage require extensive supplementation with additional readings, videos, and handouts that create a confusing, patchwork learning experience for students who must navigate multiple platforms; and that lack of integrated ancillary materials — high-quality test banks aligned to learning objectives, LMS-ready adaptive homework, and multimedia — makes formative assessment and active learning less effective and more time-consuming to manage (Jung et al., 2017). These are not arguments against OER in principle, but arguments for quality thresholds and against one-size-fits-all mandates.

Many faculty lack the time, expertise, or resources to create quality OER at the standard they hold for their teaching. Allen and Seaman (2014) found in national surveys that barriers to creation were practical: 71% of faculty cited lack of time as the primary barrier to OER creation, given existing teaching loads; 53% cited lack of technical expertise, particularly for creating accessible multimedia, interactive simulations, and professionally formatted texts; 58% cited lack of institutional support such as instructional designers, graphic designers, copyright librarians, and accessibility checkers who are standard on commercial teams; 45% cited lack of funding for development costs like software, images, or permissions; and 67% said they didn't know where to find quality OER or how to evaluate it using rubrics, indicating a major discovery and evaluation literacy gap. This means that asking faculty to "just create OER" without significant infrastructure is asking them to become author, editor, designer, and publisher overnight.

Adjunct Faculty Are Particularly Harmed:

OER mandates disproportionately burden already-exploited adjunct and contingent faculty, who now teach the majority of students. As documented in the literature on academic precarity, including in "The Precarious Professor" section, 70% or more of college courses in the U.S. are taught by contingent faculty — adjuncts, lecturers, and non-tenure-track instructors — earning poverty wages of $2,000 to $5,000 per course with no benefits, no office, and no job security. For these faculty, an OER mandate is particularly damaging. These faculty cannot afford 20 to 60 unpaid hours adapting OER when they are teaching 4 to 8 courses at multiple institutions simultaneously just for survival. They need ready-to-use, turnkey materials with robust test banks, slides, and homework systems to manage crushing workloads and last-minute course assignments, often given only days before a semester starts. They have no job security or academic freedom protections to risk resisting institutional mandates, even when they believe the OER is inadequate. And they rarely receive stipends, course releases, or instructional design support for OER work, as those resources are typically reserved for full-time faculty. Mandating that the most exploited, lowest-paid faculty absorb additional unpaid labor to solve a systemic affordability crisis created by publishers and state disinvestment is particularly egregious and reproduces inequity within the faculty itself, while further jeopardizing educational quality for the students who most need support.## Impact on Student Learning, Enrollment, and the Hidden Costs of "Free"

Learning Outcomes: Mixed Evidence with Methodological Limitations

Research on OER learning outcomes shows mostly equivalent performance on basic, short-term measures, but severe methodological limitations prevent confident conclusions about true learning. Hilton's (2016) comprehensive review of 16 OER efficacy studies, later updated to over 20, is the most cited synthesis in the field. He found that the majority of studies — 14 of 16 in the initial review — showed no significant difference in learning outcomes between students using OER and those using commercial textbooks, with two showing improvements for OER. On the surface, this "no significant difference" finding has been used to claim OER is just as good. However, Hilton himself noted severe methodological problems that limit any strong conclusion: most studies had small sample sizes and were underpowered to detect meaningful differences; virtually none used random assignment, instead comparing different semesters or instructors, introducing selection bias; there is likely strong publication bias where negative results showing OER performing worse are unpublished because institutions want to show their OER initiatives succeed; outcome measures were limited almost entirely to course grades, final exam scores, and basic recall on multiple-choice tests; higher-order thinking, conceptual understanding, transfer, and long-term retention were rarely assessed; and there was almost no long-term follow-up on whether knowledge persisted into subsequent courses or affected retention and degree completion. In short, we know OER doesn't dramatically lower grades in the same course, but we know very little about whether it produces equivalent deep learning.

Studies showing equivalent outcomes often measure only surface-level learning and course compliance, not depth of understanding. Colvard et al. (2018) conducted a large-scale study of over 20,000 students at the University of Georgia and found equivalent final grades and DFW rates with OER, a finding widely celebrated. However, the authors themselves acknowledged that course grades measure "successful navigation of course requirements" — turning in homework, participating, taking the tests the instructor designed — not necessarily depth of conceptual learning, long-term retention, or ability to apply knowledge to new contexts. Clinton and Khan (2019), in their meta-analysis of OER efficacy, note that most OER efficacy research uses course performance or final grades as the sole outcome measure, which may not capture conceptual understanding, ability to transfer knowledge, or intellectual development. Grades are also instructor-dependent and can be inflated or deflated by curving, extra credit, and changes in assessment difficulty when faculty switch materials, making them a weak proxy for learning quality.

Some emerging evidence suggests OER may disadvantage students specifically on higher-order, deeper learning outcomes, even when recall is equivalent. Croteau (2017), studying Affordable Learning Georgia courses, found that while students using OER performed equivalently on multiple-choice exams measuring recall and recognition, they scored significantly lower on essay exams and application problems requiring deeper understanding, analysis, synthesis, and transfer to new scenarios. Similarly, Grewe and Davis (2017) found that students using OER in introductory statistics courses performed equivalently on computational problems — calculating a t-test or mean — but performed significantly worse on conceptual understanding items requiring them to interpret what those statistics mean, explain assumptions, or critique a study design. This pattern — equivalent on procedural and recall tasks, weaker on conceptual and application tasks — is precisely what learning science would predict if textbooks lack rich explanations, varied worked examples, and scaffolded practice that build mental models. It suggests that OER may be "good enough" for passing the test but less effective for building the durable, flexible knowledge needed for advanced work.

Students themselves report dissatisfaction with OER quality despite appreciating cost savings, complicating the access narrative. Jhangiani and Jhangiani (2017) surveyed over 300 students and found that while students overwhelmingly appreciated cost savings and rated OER positively on cost, 42% rated OER quality — including clarity, comprehensiveness, and visual design — as lower than commercial textbooks they had used, and 38% reported needing to seek supplementary resources like YouTube, Chegg, or tutoring to understand concepts the OER did not explain well. Florida Virtual Campus (2016) surveyed over 20,000 Florida students and found that 35% of students using OER rated them as lower quality than commercial options in terms of helpfulness for learning, and many reported frustration with errors, poor formatting, and lack of study aids. Students often face a double bind: they cannot criticize free materials without seeming ungrateful, but they privately worry about being disadvantaged.

Long-term impacts on retention of knowledge, transfer to advanced coursework, and career outcomes are completely unstudied — the most critical gap in the literature. No published peer-reviewed research to date examines whether students who use lower-quality OER in foundational courses like General Chemistry I or Principles of Economics perform worse in subsequent advanced courses like Organic Chemistry II or Intermediate Microeconomics due to knowledge gaps — a critical question for understanding the true educational impact and cumulative nature of learning. Watson et al. (2017) explicitly note this research gap, stating that the field has focused almost exclusively on immediate, within-course grades and has ignored longitudinal outcomes. No studies track whether OER students require more remediation, struggle more in internships, or need more on-the-job training. Without this longitudinal research, claims that OER has no negative impact on learning are premature and based on an incomplete outcome set.

Have Students Enrolled in More Classes Due to Reduced Textbook Costs?

No evidence exists that OER adoption actually increases student course-taking or credit enrollment — a critical metric if access and affordability are the true goals, not just cost-shifting. The Florida Virtual Campus (2016) study found that while 47.6% of students self-reported that they had taken fewer courses than they wanted to due to textbook costs at some point in their college career, there is no published causal evidence that OER initiatives have actually reversed this and increased course-taking rates at an institutional level. Institutions have aggressively tracked and publicized student savings dollars — often multiplying enrollment by list price — but have not systematically tracked whether students actually enroll in more credits after OER adoption, despite this being the obvious behavioral metric if access is a genuine priority (Watson et al., 2017). If textbooks truly prevented enrollment, removing that barrier should lead to measurable increases in credits attempted and completed.

The minimal research available that does examine enrollment suggests negligible or zero enrollment effects. Colvard et al. (2018) examined enrollment patterns at the University of Georgia before and after large-scale OER adoption in eight high-enrollment courses and found no significant difference in enrollment intensity, credit loads, or subsequent semester persistence that could be attributed to OER. Course withdrawal rates show similarly small effects: Clinton and Khan's (2019) meta-analysis of 13 studies found that withdrawal rates were slightly lower with OER — a 0.5 to 1 percentage-point decrease — but effect sizes were small, confidence intervals were wide, and results were inconsistent across studies, with some showing increases. This small decrease could be due to day-one access rather than quality, as students have materials from the start.

The logic for limited enrollment impact is straightforward when considering the full economics of college attendance. Textbook costs, while significant and burdensome, are a relatively small portion of total cost of attendance — typically 2-4% when tuition, fees, housing, food, and transportation are included. Students have already developed extensive strategies to work around textbook costs, including buying used, sharing with friends, using library reserves, renting, pirating, or simply going without, which dampens the impact of making a book free. More importantly, making one or two courses free does not enable taking an additional course when tuition for those additional 3-credit courses is $1,500 to $3,000 per course at a public university and far more at a private university. And time constraints matter more than textbook costs for most working students — who are the majority of community college students — because working 30 hours per week limits how many credits they can take regardless of whether the textbook is $0 or $200. Free textbooks help cash flow but do not create more hours in the day.

The Hidden Costs Students Actually Pay:

While OER eliminates upfront textbook costs, students may pay far more in hidden, downstream costs and long-term consequences that are not captured in savings calculations. These hidden costs include: needing to purchase additional materials or resources to supplement inadequate OER — such as commercial study guides, test prep books, Chegg subscriptions, private tutoring, or supplemental online courses — directly defeating the "free" purpose; spending significantly more time and cognitive effort teaching themselves content that should have been clearly explained with worked examples and scaffolding in a high-quality text; struggling academically due to poor or incomplete materials, leading to lower grades, need to retake courses, extended time to degree, and additional tuition; lacking adequate preparation for advanced courses, internships, research experiences, or jobs, requiring remediation or additional training that employers and graduate schools must provide; performing poorly on standardized professional licensure exams like NCLEX for nursing, FE for engineering, or CPA exams, requiring expensive retakes and delaying entry into the profession; and missing out on the learning, intellectual development, and curiosity that quality materials with rich examples, case studies, and diverse perspectives would have provided — an opportunity cost that affects career trajectory, graduate school options, and lifetime earning potential in ways that are difficult to quantify but very real.

No research has comprehensively examined these total, true costs to students over time. Watson et al. (2017) note that while OER efficacy research measures immediate course outcomes within a single semester, no studies examine long-term impacts on "career preparedness, professional licensure exam pass rates, graduate school admissions and success, or lifetime earnings" — all of which are potentially and plausibly affected by foundational knowledge gaps from lower-quality OER in introductory courses. Institutions publish savings figures based on list price multiplied by enrollment, but no institution publishes a true cost-benefit analysis that subtracts the costs of faculty time, student time spent self-teaching, tutoring costs, or downstream impacts on retention in the major. The false economy of "free but inadequate" materials may cost students tens of thousands more in career earnings, extended education, and lost opportunities than $1,000 in textbook costs would have, but that economy is invisible because we do not measure it.

The Self-Teaching Burden Falls on Students:

When OER materials are inadequate, the instructional burden does not disappear — it is transferred to students, who are left to teach themselves without expert guidance. They must: identify gaps in their own understanding without a clear framework for knowing what they don't know — a metacognitive task even experts struggle with; search for supplementary materials on the open web without guidance on quality, accuracy, or appropriateness for their course level; teach themselves complex, threshold concepts using random YouTube videos, websites, forums like Reddit, or AI tools of unknown quality and accuracy that may contain misconceptions; spend extensive additional time — often 10 to 20+ hours per course as documented in student surveys — finding and learning from these supplementary materials; and essentially pay tuition to self-teach while faculty who assigned inadequate materials may not acknowledge or address the gaps, because they believe the OER is sufficient.

This dynamic creates enormous inequity and undermines the equity rationale for OER. Students with prior knowledge from well-resourced high schools, strong study skills, access to private tutoring, high-speed internet and quiet study spaces, and time to self-teach can supplement successfully and may even thrive with OER. First-generation students, working students with jobs and caregiving responsibilities, students without prior preparation in the subject, and those lacking time, academic confidence, or resources struggle disproportionately because they depend more heavily on high-quality, comprehensive, self-contained course materials. When those materials are absent, the gap widens. In this way, poorly implemented OER becomes another mechanism that advantages already-advantaged students while disadvantaging those it was explicitly meant to help, reproducing stratification under the banner of affordability.

Student Choice: Should Students Decide Between OER and Commercial Textbooks?

The topic of letting students choose between commercial and OER textbooks does not often come up in OER research, but it raises important issues such as autonomy, fairness, and learning styles. Right now, faculty or, sometimes, administrators pick the materials for entire courses, so every student uses the same resources. What if students could decide for themselves whether to buy a commercial textbook or use a free OER option?

Arguments for Student Choice:

- Respects student autonomy and diverse learning needs. Students are adults paying (often through loans) for education; they should have input into learning materials. Jhangiani and Jhangiani (2017) found that students value having options, and some students prefer print textbooks they can annotate, keep as references, and use without screens.
- Addresses equity concerns both ways. Student choice allows: students who cannot afford textbooks to use free OER without shame or stigma, AND students who can afford textbooks to purchase higher-quality materials if they judge OER inadequate. This avoids forcing all students to use potentially inferior materials to address the financial needs of some students.
- Enables natural quality comparison and feedback. If students could choose, their choices would provide valuable data about perceived quality, usefulness, and value. If most students chose commercial textbooks despite the cost, this would signal concerns about OER quality. If most chose OER, this would validate its adequacy.
- Reduces faculty burden and conflict. Rather than faculty having to choose between student finances and educational quality, students make their own cost-quality trade-offs based on individual circumstances, learning preferences, and priorities.

Arguments Against Student Choice (or Why This Isn't Implemented):

- Creates two-tiered learning experiences within same course. Students using different materials might have different depths of coverage and understanding, access different practice problems and resources, be prepared differently for exams, and experience different cognitive load and learning efficiency. Faculty would need to accommodate multiple materials, teach using multiple resources, and create assessments that are fair to both groups—significantly increasing the workload.
- Disadvantages students who choose free option. If OER is genuinely inferior (as research suggests it often is), students choosing it due to financial necessity would receive a worse education than wealthier peers—making the cost barrier visible and explicit within the classroom rather than hidden. This could create shame, resentment, and documented educational inequality.
- Complicates instruction and assessment significantly. Faculty couldn't reference specific textbook pages, examples, or problems if students have different materials. Homework from textbook problem sets wouldn't work if some students lack access. Lecture alignment with textbook would benefit only some students.

What Research Reveals About Student Preferences:

- Limited research exists because choice is rarely offered, but studies reveal interesting patterns. Jhangiani and Jhangiani (2017) found that when asked hypothetically: 67% of students said they would choose free OER over $150 commercial textbook, but 42% of those same students rated OER quality lower than commercial textbooks they'd used previously, and 31% said they would purchase commercial textbook if they could afford it and believed it was significantly better quality.
- This suggests students make cost-quality trade-offs, not just cost minimization. Many students would pay for textbooks if they believed the quality difference was meaningful and they could afford it. The fact that textbooks cost $200-$400 (unconscionable amounts for many students) drives OER adoption, not that students believe free is always optimal.

What Would Happen If Student Choice Was Implemented?

Impacts on Universities:

- Would expose whether OER adoption is truly about student benefit or cost-cutting. If institutions claim OER is equal quality and students prefer it, offering choice should result in universal OER adoption. If students systematically choose commercial textbooks when given option, this reveals quality concerns and that institutions have been forcing inferior materials.
- Could increase administrative complexity and costs. Universities would need to: stock both OER and commercial textbooks (or provide access to both), manage two sets of materials in course reserves, accommodate different student materials in learning management systems, and track which students use which materials.
- Might reduce OER adoption if quality concerns are real. If significant numbers of students choose commercial textbooks, faculty might feel validated in quality concerns and push back against OER mandates, potentially derailing institutional OER initiatives and associated PR benefits.
- Would make cost-quality trade-offs explicit and potentially embarrassing. Media coverage of "students forced to choose between rent and textbooks" or "wealthy students get better education in same course" could create negative publicity institutions want to avoid.

Impacts on Professors:

- Would increase instructional workload significantly. Faculty would need to: accommodate multiple materials in course design, create assessments that are fair to students using different resources, field questions about both sets of materials, stay current with both OER and commercial options, and potentially teach the same content differently for different groups.

- Could validate faculty expertise and professional judgment. If student choices align with faculty quality assessments (students choosing commercial textbooks in courses where faculty believe OER is inadequate), this validates faculty professional judgment and could strengthen arguments for faculty autonomy.
- Would require communicating trade-offs to students. Faculty would need to advise students on which option to choose, explaining the strengths/weaknesses of each—time-consuming, potentially uncomfortable conversations about quality differences.

Why Student Choice Isn't Widely Implemented—The Uncomfortable Truth:

- Student choice would reveal quality differences institutions prefer to obscure. If OER is genuinely equal quality, why not offer choice? The fact that choice is rarely offered suggests that institutions know many students would choose commercial textbooks if they were affordable, revealing that "free" OER involves quality trade-offs that institutions don't want explicitly acknowledged.
- Universities benefit from mandatory OER because it reduces costs and maximizes PR claims. If only some students use OER, cost savings are reduced and headlines about "free textbooks for all students" become "free textbooks for some students," weakening marketing value.
- The current model allows institutions to control narrative and metrics. With student choice, students might choose differently than institutions prefer, complicating reporting and assessment.

When OER Works Well and Better Solutions

Conditions for OER Success:

High-quality OER does exist and can equal or exceed commercial textbooks in both student satisfaction and learning outcomes, but only under specific, resource-intensive conditions that are rarely met by default. Research synthesizing successful OER implementations identifies a consistent set of success factors that mirror commercial publishing infrastructure rather than volunteerism. These include: adequate, sustained funding for professional development and project management rather than one-time stipends (Pitt et al., 2020); teams of discipline experts co-authoring and dividing labor by specialty rather than single individual authors writing entire textbooks outside their subspecialty (Allen et al., 2015); rigorous, multi-stage peer review processes involving content experts, pedagogical reviewers, and student reviewers (Jung et al., 2017); professional editing, design, illustration, and accessibility remediation to meet WCAG standards, rather than faculty formatting in Word (Pawlyshyn et al., 2013); regular updating with sustainable, funded maintenance models that pay for new editions, errata, and link-checking rather than assuming volunteers will maintain it indefinitely (Petrides et al., 2011); comprehensive ancillary materials including test banks, slides, homework systems, and instructor guides that make adoption manageable (Delimont et al., 2016); faculty choice and professional autonomy rather than top-down mandates, so adoption is driven by pedagogical fit (Belikov & Bodily, 2016); and adequate institutional support including course release time, instructional designers, librarians, and recognition in promotion and tenure (Henson, 2017). When all of these are present, OER succeeds; when most are absent, quality suffers.

OpenStax represents the best-case scenario and the clearest proof that this model can work, but it also illustrates why most OER cannot replicate it without similar investment. OpenStax has received over $20 million in foundation funding from Hewlett, Gates, Arnold Ventures, and others, plus ongoing revenue from low-cost print sales and institutional partnerships. It uses that funding to employ teams of 10-20 expert authors per book, 50+ peer reviewers per book, full-time PhD content producers, professional editors, designers, accessibility experts, and error-tracking systems, with ongoing maintenance and regular new editions — essentially replicating a commercial publisher's infrastructure but with an open license. The result is

OER that faculty consistently rate as comparable to commercial texts. However, as Pitt (2015) and others note, most OER lacks these resources by an order of magnitude. A typical OER grant of $1,000 to $3,000 to a single faculty member cannot produce OpenStax-level quality, ancillaries, or sustainability, leading to a two-tiered system where a few flagship OER are excellent and the long tail is highly variable. Scaling OpenStax quality to cover the entire curriculum would require hundreds of millions in sustained funding that does not currently exist.

Alternative Models That Balance Quality, Cost, and Choice:

Universities negotiate institutional bulk rates with commercial publishers for digital access to the exact same vetted, peer-reviewed commercial textbooks, homework platforms, and ancillaries, which are then distributed to all students on the first day of class via the LMS at a sharply reduced cost — typically $25 to $50 per textbook instead of $200 to $400 list price — and charged as a course materials fee. Students are automatically enrolled but can opt out if they prefer to source OER, used books, or other alternatives. Research by Cummings et al. (2019) and data from publishers and institutions suggests this model has several measurable advantages: it significantly reduces costs by 60-75% compared to new print while preserving publisher revenue enough to sustain quality infrastructure; it maintains quality by using fully vetted commercial materials with full ancillary support, which faculty already trust and know how to use; it provides universal, equitable day-one access so all students have required materials from the first assignment, eliminating the 20-30% of students who previously delayed or avoided purchase; and it reduces piracy and unauthorized sharing that creates legal risk.

However, concerns have been raised by student advocates and OER advocates: students are charged automatically without always seeing the cost clearly during registration, which reduces price transparency compared to seeing a textbook price at a bookstore; opting out often requires extra effort, navigating forms and deadlines, and students may lose access to homework platforms if they opt out, creating a default enrollment nudge; publishers still profit and retain market power while potentially undermining the used textbook market, library reserves, and OER alternatives that could be even cheaper; and some faculty worry that inclusive access entrenches publisher platforms and data collection. Despite these concerns, for institutions whose primary goal is immediate affordability with minimal quality risk and minimal faculty workload increase, inclusive access has become the fastest-growing affordability model, now surpassing pure OER in number of students reached.

Some institutions, notably community college consortia like the Open Course Library in Washington, the Virginia Community College System, and Lumen Learning's co-published courses, have invested not in mandating OER but in vetting high-quality OER, providing faculty-curated collections that have been reviewed for quality, accuracy, and accessibility using standardized rubrics, and supplementing them with librarian, instructional designer, and accessibility support for faculty who choose them. This model directly addresses quality concerns while maintaining affordability. Instead of asking each faculty member to individually find and vet OER — requiring 20-40 hours of work — a central team of librarians and faculty fellows does the vetting once, creates detailed reviews, maps OER to learning outcomes, and provides implementation guides. Faculty retain full choice: they can adopt the vetted OER, adapt it, or choose commercial. Early evidence suggests this curated model produces higher faculty satisfaction and higher-quality implementations than pure mandates, because it respects professional judgment while lowering the discovery burden.

Some institutions provide all students with grants or vouchers of $200 to $500 per semester to purchase required textbooks — whether commercial or OER, with the choice left to the student or faculty — funded by institutional subsidy, student fees, or reallocated bookstore commissions. This respects student autonomy and faculty academic freedom, ensures access regardless of financial need because funds are provided upfront, and allows market forces and transparent quality preferences to emerge: if OER is truly better, students and faculty will choose it when given equal purchasing power. It also avoids cost-shifting to unpaid faculty labor, because the institution bears the cost directly. However, this model requires actual, sustained institutional financial commitment rather than cost-shifting rhetoric. It requires universities to allocate real dollars from tuition, fees, endowment, or state appropriations to cover materials, rather than simply telling faculty to create free materials so the institution can claim savings while continuing to raise tuition and administrative spending. For institutions unwilling to make that commitment, grants remain rare, but they represent the most equitable model from a student autonomy perspective.

What Would Actually Help:

Address the root cause of high commercial textbook prices through direct consumer protection and market regulation, rather than placing the entire burden on faculty to work around publishers. Senack (2016) and student PIRG reports have long documented textbook market failures that justify policy intervention: publishers exploit a captive market where faculty assign but students pay, eliminating normal price sensitivity. Effective interventions would include: capping prices or price increases for required course materials when institutions have exclusive contracts; prohibiting forced new editions without substantial, pedagogically justified changes — currently publishers often release new editions every 2-3 years with minimal changes simply to kill the used book market; unbundling access codes from textbooks so students are not forced to buy a new $200 book to get a $50 homework code that expires; requiring that publishers make affordable used textbook, rental, and e-book options available and not use digital rights management to block resale; and enforcing anti-trust enforcement against publisher consolidation, where the market has shrunk from dozens of competitors to three major oligopolists — Pearson, Cengage/McGraw-Hill — which reduces competition. These reforms would preserve quality and choice while directly addressing profiteering, without relying on volunteer faculty labor to create a parallel free system from scratch.

Address the root cause of high commercial textbook prices through direct consumer protection and market regulation, rather than placing the entire burden on faculty to work around publishers. Senack (2016) and student PIRG reports have long documented textbook market failures that justify policy intervention: publishers exploit a captive market where faculty assign but students pay, eliminating normal price sensitivity. Effective interventions would include: capping prices or price increases for required course materials when institutions have exclusive contracts; prohibiting forced new editions without substantial, pedagogically justified changes — currently publishers often release new editions every 2-3 years with minimal changes simply to kill the used book market; unbundling access codes from textbooks so students are not forced to buy a new $200 book to get a $50 homework code that expires; requiring that publishers make affordable used textbook, rental, and e-book options available and not use digital rights management to block resale; and enforcing anti-trust enforcement against publisher consolidation, where the market has shrunk from dozens of competitors to three major oligopolists — Pearson, Cengage/McGraw-Hill — which reduces competition. These reforms would preserve quality and choice while directly addressing profiteering, without relying on volunteer faculty labor to create a parallel free system from scratch.

Restore and protect faculty professional judgment and autonomy to choose the best materials — OER, commercial, or hybrid — based on pedagogical needs, disciplinary expertise, and quality rather than cost metrics alone. Effective policy would allow faculty to choose the best materials for their students based on quality and learning outcomes, not on whether they are free. It would provide meaningful stipends and release time to faculty who create supplementary materials for OER — test banks, homework sets, slides, videos — recognizing that creating ancillaries is real labor essential for quality. And it would respect professional judgment about which materials serve students best in each specific course context (Belikov & Bodily, 2016). This means moving away from soft mandates, departmental savings quotas, and public dashboards that shame faculty who choose commercial texts for pedagogical reasons, and toward a model where OER is a supported, incentivized option among many, not a required default. Faculty who know their students and disciplines best should be trusted partners in affordability work, not targets of pressure campaigns.

Actually reduce the overall cost of higher education through structural reinvestment, rather than using textbook affordability as a distraction from tuition inflation. The focus on textbook costs, while important, can obscure the larger affordability crisis. True affordability requires state reinvestment in public higher education, which has declined by over 30% per student in real terms since 2000, shifting costs directly onto students via tuition (SHEEO, 2020). It requires institutions to reduce administrative bloat and executive compensation growth, which has far outpaced instructional spending. It requires prioritizing instructional spending — faculty, advisors, libraries, labs — over amenities, marketing, and athletics that drive tuition increases. And it requires making higher education genuinely affordable through need-based aid, tuition caps tied to inflation, and funding models that reward completion, not just enrollment. Textbook costs are a symptom of a marketized higher education system where costs are passed to students at every turn; OER alone cannot solve that systemic problem. If institutions can find $100 million for a new stadium or administration building, they can find funds to ensure all students have high-quality materials without exploiting faculty volunteer labor. Real affordability means the institution absorbs costs rather than shifting them.

The Bottom Line

Research on Open Educational Resources (OER) reveals a complex situation that requires thoughtful attention. The affordability crisis in commercial textbooks is well known, and there is evidence of publisher exploitation (GAO, 2013; Florida Virtual Campus, 2016). However, when institutions require OER adoption, it often seems driven more by public relations, cost savings, and political considerations than by evidence of better educational outcomes (Henson, 2017; Belikov & Bodily, 2016).

Several major repositories make up the OER landscape. OpenStax has reached 3.6 million students in over 14,000 courses (OpenStax, 2022). The BC Campus Open Collection has been adopted in more than 1,000 courses and saved students $20 million (BCcampus, 2021). The Open Textbook Library offers over 950 textbooks, and LibreTexts has seen more than 1 billion accesses and over 12,000 course adoptions (Delmas et al., 2022). The AIM Open Textbook Initiative features 90 approved math textbooks. Even with these resources, OER is used in only about 8-10% of higher education courses (Seaman & Seaman, 2021). Adoption rates vary: 15-30% in lower-division general education courses at institutions with OER programs, 5-10% in upper-division courses, and 2-5% in specialized or graduate courses. Community colleges use OER more (12-15%) than four-year schools (6-9%), and states with funding see adoption rates two to three times higher than those without. While use has doubled from 3-4% in 2016 to 8-10% in 2021, most courses (over 90%) still use commercial textbooks. This may be due to a lack of high-quality OER for many courses, or to faculty concerns about whether available OER meets their needs.

OER is mostly used in lower-division general education courses, where commercial textbooks are the most expensive and attract the most attention. In contrast, commercial materials still dominate upper-division and specialized courses. This suggests that institutions may be focusing on reducing visible costs rather than on broad improvements in educational quality. Studies have found several quality issues: only 23% of OER materials are peer-reviewed (Jung et al., 2017), there are gaps in coverage and rigor (Bliss et al., 2013), 67% of faculty say there are not enough supplementary materials (Jung et al., 2017), 40-60% of OER is more than five years old (Pitt, 2015), and there are not enough practice problems in STEM subjects (Winitzky-Stephens & Pickavance, 2017).

Universities benefit from OER mandates by getting good publicity, cutting costs, and gaining political support, but they also shift unpaid work onto faculty. This unpaid labor is estimated at $15,000- $30,000 per course (Petrides et al., 2011), which can undermine student learning. Many faculty feel pressured (63%; Belikov & Bodily, 2016), worry about quality (47%), and dislike the 20-60 hours of unpaid work needed for each course (Delimont et al., 2016). Research shows that student performance on basic measures is similar, but the studies have important limitations (Hilton, 2016). Some findings show worse results in higher-order thinking skills (Croteau, 2017; Grewe & Davis, 2017) and that 42% of students rate OER as lower quality (Jhangiani & Jhangiani, 2017). There is also no evidence that OER adoption increases enrollment, which would be important if access was the main goal (Watson et al., 2017).

There are better ways to address textbook costs, such as regulating commercial publishers (Senack, 2016), investing in high-quality OER with proper review and upkeep (like the OpenStax model with enough funding; Allen et al., 2015; Pitt et al., 2020), protecting faculty choice (Belikov & Bodily, 2016), trying inclusive access or subsidy models, and most importantly, increasing public investment to tackle the root causes (SHEEO, 2020). Students should have both affordable and high-quality materials, not one or the other. Current adoption rates—8-10% overall, mostly in lower-division courses, with over 90% of courses still using commercial materials—show that OER often acts as a symbolic fix, saving only 2-4% of total attendance costs, while tuition keeps rising (making up 50-70% of total costs and increasing by 20-35% in the same period). The "hidden cost of free" falls on students through lower educational quality, on faculty through unpaid work, and on society by producing graduates who are less prepared for their careers. This allows institutions to appear to be addressing affordability issues without making meaningful changes to their priorities.

References

Allen, G., Guzman-Alvarez, A., Molinaro, M., & Larsen, D. (2015). Assessing the impact and efficacy of the open-access ChemWiki textbook project. Educause Learning Initiative Brief. https://library.educause.edu/resources/2015/1/assessing-the-impact-and-efficacy-of-the-openaccess-chemwiki-textbook-project

Becker, Kristen L., et al. "High-Priced Textbooks' Impact on Community College Student Success." Community College Journal of Research and Practice, vol. 47, no. 2, 2023, pp. 1-20. https://doi.org/10.1177/00915521221125898

"Open Textbooks Pilot Program." U.S. Department of Education, 2024. https://www.ed.gov/grants-and-programs/grants-higher-education/improvement-of-postsecondary-education/open-textbooks-pilot-program-84116t Accessed August 8, 2026

Walsh, Andrew. "Removing barriers: textbook affordability and OER at Sinclair community college." Reference Services Review ahead-of-print, 2020. https://doi.org/10.1108/RSR-03-2020-0011

Walsh, Andrew. "Removing barriers: textbook affordability and OER at Sinclair community college." Reference Services Review, 2020. https://doi.org/10.1108/RSR-03-2020-0011

Herbert, Michael J., et al. "Faculty Motivation for OER Textbook Adoption and Future Use." Innovative Higher Education, vol. 48, 2022. https://doi.org/10.1007/s10755-022-09625-6

Becker, Andrea, et al. "Shared Governance and Systems Theory: A Mixed Methods Study of Faculty Perceptions and Ideas." Higher Education Politics and Economics, vol. 9, no. 2, 2021, pp. 1-20. https://doi.org/10.32674/hepe.v9i2.5974

Herbert, Michael J., et al. "Faculty Motivation for OER Textbook Adoption and Future Use." Innovative Higher Education, vol. 48, no. 2, 2022. https://doi.org/10.1007/s10755-022-09625-6

Orzech, Mary Jo, et al. "All about OER – Why, What, How, and So What: A Collective Case Study." The Journal of Higher Education, vol. 96, no. 1, 2025, pp. 1-20. https://doi.org/10.1177/00472395241309876

Allen, I. E., & Seaman, J. (2014). Opening the curriculum: Open educational resources in U.S. higher education, 2014. Babson Survey Research Group. https://www.onlinelearningsurvey.com/reports/openingthecurriculum2014.pdf

BCcampus. (2021). Open education annual report 2020-2021. https://bccampus.ca/open-education-annual-report-2020-2021/

Belikov, O., & Bodily, R. (2016). Incentives and barriers to OER adoption: A qualitative analysis of faculty perceptions. Open Praxis, 8(3), 235-246. https://doi.org/10.5944/openpraxis.8.3.308

Bliss, T. J., Hilton, J., Wiley, D., & Thanos, K. (2013). The cost and quality of online open textbooks: Perceptions of community college faculty and students. First Monday, 18(1). https://doi.org/10.5210/fm.v18i1.3972

Clements, K. I., & Pawlowski, J. M. (2012). User-oriented quality for OER: Understanding teachers' views on re-use, quality, and trust. Journal of Computer Assisted Learning, 28(1), 4-14. https://doi.org/10.1111/j.1365-2729.2011.00450.x

Clinton, V., & Khan, S. (2019). Efficacy of open textbook adoption on learning performance and course withdrawal rates: A meta-analysis. AERA Open, 5(3). https://doi.org/10.1177/2332858419872212

College Board. (2021). Trends in college pricing and student aid 2021. https://research.collegeboard.org/trends/college-pricing

Colvard, N. B., Watson, C. E., & Park, H. (2018). The impact of open educational resources on various student success metrics. International Journal of Teaching and Learning in Higher Education, 30(2), 262-276.

Croteau, E. (2017). Measures of student success with textbook transformations: The Affordable Learning Georgia initiative. Open Praxis, 9(1), 93-108. https://doi.org/10.5944/openpraxis.9.1.505

Cummings, L., Rodriguez, J., & Crosslin, M. (2019). Affordable textbook options in higher education. Journal of College Access, 4(2), 35-49.

Delmas, P., Read, K., Becker, B., & Larkin, K. (2022). The LibreTexts libraries: Open access textbooks in an open educational ecosystem. Open Praxis, 14(1), 12-29. https://doi.org/10.55982/openpraxis.14.1.386

Delimont, N., Turtle, E. C., Bennett, A., Adhikari, K., & Lindshield, B. L. (2016). Perceived barriers to the development of open educational resources: A survey of educators. The International Review of Research in Open and Distributed Learning, 17(1). https://doi.org/10.19173/irrodl.v17i1.2150

Feldstein, A., Martin, M., Hudson, A., Warren, K., Hilton, J., & Wiley, D. (2012). Open textbooks and increased student access and outcomes. European Journal of Open, Distance and E-Learning. http://www.eurodl.org/?p=current&article=533

Fischer, L., Hilton, J., Robinson, T. J., & Wiley, D. A. (2015). A multi-institutional study of the impact of open textbook adoption on the learning outcomes of post-secondary students. Journal of Computing in Higher Education, 27(3), 159-172. https://doi.org/10.1007/s12528-015-9101-x

Florida Virtual Campus. (2016). 2016 student textbook and course materials survey. https://dlss.flvc.org/documents/210036/1314923/2016+Student+Textbook+Survey.pdf

Florida Virtual Campus. (2021). 2020-2021 OER and affordable course materials in Florida report. https://dlss.flvc.org/oer

Ginsberg, B. (2011). The fall of the faculty: The rise of the all-administrative university and why it matters. Oxford University Press.

Grewe, K., & Davis, W. P. (2017). The impact of enrollment in an OER course on student learning outcomes. The International Review of Research in Open and Distributed Learning, 18(4). https://doi.org/10.19173/irrodl.v18i4.2986

Henson, L. (2017). Exploring faculty perceptions of administrative support for open educational resources. Journal of Open, Flexible and Distance Learning, 21(1), 27-43.

Hilton, J. (2016). Open educational resources and college textbook choices: A review of research on efficacy and perceptions. Educational Technology Research and Development, 64(4), 573-590. https://doi.org/10.1007/s11423-016-9434-9

Hilton, J., Robinson, T. J., Wiley, D., & Ackerman, J. D. (2014). Cost-savings achieved in two semesters through the adoption of open educational resources. The International Review of Research in Open and Distributed Learning, 15(2). https://doi.org/10.19173/irrodl.v15i2.1700

Jhangiani, R. S., & Jhangiani, S. (2017). Investigating the perceptions, use, and impact of open textbooks: A survey of post-secondary students in British Columbia. The International Review of Research in Open and Distributed Learning, 18(4). https://doi.org/10.19173/irrodl.v18i4.3012

Jung, E., Bauer, C., & Heaps, A. (2017). Higher education faculty perceptions of open textbook adoption. The International Review of Research in Open and Distributed Learning, 18(4). https://doi.org/10.19173/irrodl.v18i4.3120

Kimmons, R. (2015). OER quality and adaptation in K-12: Comparing teacher evaluations of copyright-restricted, open, and open/adapted textbooks. The International Review of Research in Open and Distributed Learning, 16(5). https://doi.org/10.19173/irrodl.v16i5.2341

OpenStax. (2022). Impact report 2022. https://openstax.org/impact

Ozdemir, O., & Hendricks, C. (2017). Instructor and student experiences with open textbooks, from the California open online library for education (Cool4Ed). Journal of Computing in Higher Education, 29(1), 98-113. https://doi.org/10.1007/s12528-017-9138-0

Pawlyshyn, N., Braddlee, Casper, L., & Miller, H. (2013). Adopting OER: A case study of cross-institutional collaboration and innovation. Educause Review. https://er.educause.edu/articles/2013/11/adopting-oer-a-case-study-of-crossinstitutional-collaboration-and-innovation

Petrides, L., Jimes, C., Middleton-Detzner, C., Walling, J., & Weiss, S. (2011). Open textbook adoption and use: Implications for teachers and learners. Open Learning: The Journal of Open, Distance and e-Learning, 26(1), 39-49. https://doi.org/10.1080/02680513.2011.538563

Pitt, R. (2015). Mainstreaming open textbooks: Educator perspectives on the impact of OpenStax college open textbooks. The International Review of Research in Open and Distributed Learning, 16(4). https://doi.org/10.19173/irrodl.v16i4.2381

Pitt, R., Ebrahimi, N., McAndrew, P., & Coughlan, T. (2020). Assessing OER quality: A mixed-methods approach. The International Review of Research in Open and Distributed Learning, 21(2), 81-93. https://doi.org/10.19173/irrodl.v21i2.4531

Seaman, J. E., & Seaman, J. (2021). Freeing the textbook: Educational resources in U.S. higher education, 2021. Bay View Analytics. https://www.bayviewanalytics.com/reports/freeingthetextbook2021.pdf

Senack, E. (2016). Covering the cost: Why we can no longer afford to ignore high textbook prices. The Student PIRGs. https://studentpirgs.org/2016/02/03/covering-cost/

SPARC (Scholarly Publishing and Academic Resources Coalition). (2021). OER state policy playbook. https://sparcopen.org/our-work/state-policy-tracking/

State Higher Education Executive Officers Association (SHEEO). (2020). State higher education finance: FY 2020. https://sheeo.org/project/shef/

U.S. Government Accountability Office. (2013). College textbooks: Students have greater access to textbook information (GAO-13-368). https://www.gao.gov/products/gao-13-368

Watson, C. E., Domizi, D. P., & Clouser, S. A. (2017). Student and faculty perceptions of OpenStax in high enrollment courses. The International Review of Research in Open and Distributed Learning, 18(5). https://doi.org/10.19173/irrodl.v18i5.2462

Winitzky-Stephens, J. R., & Pickavance, J. (2017). Open educational resources and student course outcomes: A multilevel analysis. The International Review of Research in Open and Distributed Learning, 18(4).

https://doi.org/10.19173/irrodl.v18i4.3118